

Magnetic Reconnection as a Potential Driver of Non-Thermal Particle Emission in Astrophysical Plasmas



Emily Simon

Advisors: Luca Comisso & Lorenzo Sironi

INTRODUCTION

Events like stellar flares are capable of producing a large fraction of high-energy particles. This phenomenon is still not fully understood. We use simulations to test if magnetic reconnection is capable of producing the observed non-thermal particle energy spectra.

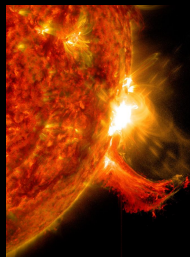
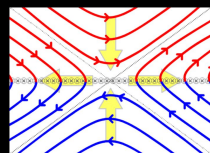
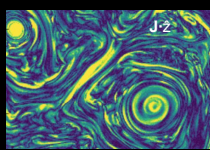


Photo credit: NASA/SDO

BACKGROUND

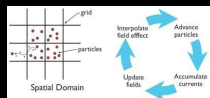


Magnetic reconnection occurs when oppositely directed magnetic field lines undergo a topological reconfiguration releasing magnetic energy.



Current sheets are sheet-like tangential field discontinuities and natural locations for magnetic reconnection to occur.

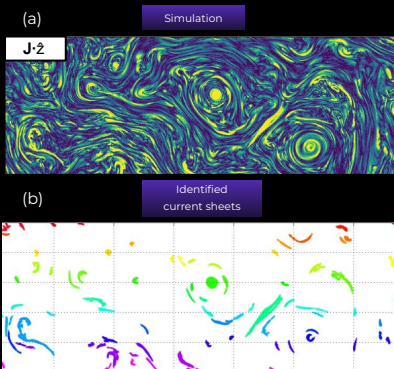
Our approach:
Particle in cell (PIC) simulations have a grid-like structure of cells in which particles move subject to the Lorentz force.



METHODOLOGY

IDENTIFYING CURRENT SHEETS:

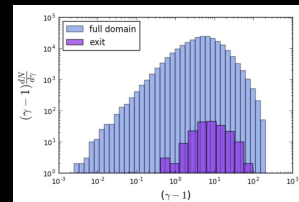
High currents (yellow) in the out of plane direction signal a possible reconnecting current sheet (Fig. a). Using a density-based clustering algorithm, each current sheet is identified & assigned a unique label (Fig. b).



EVOLVING CURRENT SHEETS OVER TIME:

In a turbulent system, we need to be able to track a current sheet during its lifetime. This is challenging because current sheets form spontaneously, change over time, and eventually die.

PARTICLE ENERGY SPECTRA

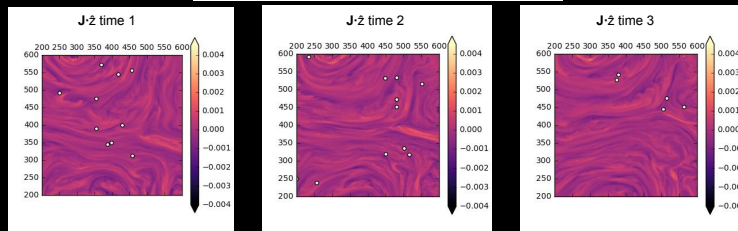


Particle energy spectra is taken over the full domain (blue) and exclusively from particles exiting a reconnecting current sheet within the domain (purple).

This preliminary work shows a large distribution of particles with higher energies after interacting with a reconnecting current sheet. This is a promising result, but more complete analysis still needs to be done.

PARTICLE ANALYSIS: Once we identify and track a current sheet throughout its lifetime, we can study the particles that have interacted with it. We take the particle energy spectrum of all particles that have just exited the current sheet and look for evidence of non-thermal acceleration.

Particles and fluid elements both evolve in time



CONCLUSION

In the near future, we look forward to improvements in our ability to verify and track reconnecting current sheets and a corresponding improvement in our particle energy spectra.

This will lead to a greater understanding of the physical mechanisms behind high-energy particle production in many different systems.

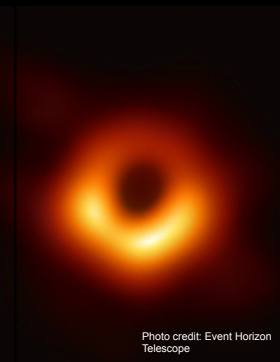


Photo credit: Event Horizon Telescope